PROJECT PRESENTATION & DEFENSE

CardioTwin X

Adaptive Simulation-Driven Cardiovascular Digital Twin

- Real-Time Non-Invasive Hemodynamic Assimilation & State Estimation
- Mechanistic 3-Element Windkessel Modeling + Unscented Kalman Filtering
- Mahalanobis Innovation Divergence Engine & 50-Twin Monte Carlo Ensemble
- Interactive 3D Biomechanical Cardiac Twin & Counterfactual What-If Lab
- Team: Kanchana A, Sinchana, Bhanushree M, Bhuvana H R | BMSIT&M

INSTITUTIONAL METADATA

Project Details & Institutional Affiliation

Academic, Engineering & Open-Source Overview

- Project Title: CardioTwin X: Adaptive Simulation-Driven Cardiovascular Digital Twin
- Institution: BMS Institute of Technology and Management (BMSIT&M), Bengaluru - 560119
- Affiliation: Visvesvaraya Technological University (VTU) | Approved by AICTE, New Delhi
- Department: Department of Computer Science and Engineering
- Team Members: Kanchana A, Sinchana (USN: 1TE25CS281), Bhanushree M, Bhuvana H R
- Repository: https://github.com/kanchanadoddaballapur-a11y/CardioTwin-X
- License: Open-Source MIT License (Permissive, Research & Educational Accessibility)

THE CHALLENGE

Problem Statement & Clinical Motivation

Addressing the Blind Spot in Modern Hemodynamic Telemetry

- Cardiovascular diseases (CVDs) cause 17.9M global deaths annually, driven by acute events.
- Standard Telemetry Limitations: Cuff SBP/DBP/HR measurements are intermittent and surface-level.
- The Hidden Physiology: Key internal parameters (Vascular Resistance Rp, Arterial Compliance C, Stroke Volume SV) are clinically unobservable without invasive catheterization.
- Lack of Mechanistic Reasoning: Black-box neural networks lack physiological laws and fail under sensor dropouts or motion artifacts.
- The Solution: A physics-grounded mathematical twin that assimilates continuous non-invasive telemetry into continuous internal hemodynamic estimates in real time.

CLINICAL TRANSLATION

Healthcare Use Cases & Clinical Impact

Transforming Continuous Patient Telemetry into Actionable Insight

1. Continuous ICU & Step-Down Monitoring: Real-time tracking of vascular resistance and compliance shifts prior to blood pressure decompensation.

2. Early Anomaly Detection: Immediate alerting for hypertensive crisis, acute heart failure, hypovolemia, and septic shock vasodilation.

3. Remote Patient Monitoring (RPM): Ingesting wearable vitals for chronic hypertension and elderly cardiac surveillance.

4. What-If Counterfactual Lab: Simulating interventions (vasodilators, beta-blockers, fluid bolus, exercise) on patient-specific physiology before clinical execution.

5. Clinician Decision Support: Plain-English explanations of parameter shifts (z-score, sigma metrics) and continuous sensor data reliability scoring.

SYSTEM ARCHITECTURE

Technical Architecture & Data Pipeline

Four Interconnected Layers of Ingestion, Estimation, Divergence & UI

- Ingestion Layer: 24h streaming telemetry with circadian rhythm, exercise episodes, noise, and simulated dropouts.
- Estimation Core: Unscented Kalman Filter (UKF) propagating 9 deterministic sigma-points through 3-Element Windkessel ODEs.
- Innovation Divergence Engine: Mahalanobis Distance D_M tracking deviations between predicted vs observed vitals.
- Explainability & Timeline: Plain-English parameter shift narratives with confidence metrics and 24h event timeline.
- Dynamic Visualization: Plotly 3D anatomical heart mesh with pressure-mapped textures and contraction dynamics.

MATHEMATICAL FOUNDATIONS

AI/ML Models & Physiological Framework

Physics-Informed Hemodynamic Mechanics + Statistical Learning

- 3-Element Windkessel (WK3): dPc/dt = (Qin(t) - Pc/Rp) / C, Pao(t) = Pc(t) + Rc*Qin(t). Solved with SciPy RK45.
- Baroreflex Feedback: Autonomous heart rate regulation loop coupled to Mean Arterial Pressure (MAP setpoint 93.3 mmHg).
- Unscented Kalman Filter (UKF): Formulated from scratch in NumPy; tracks state x = [Rp, C, SV, HR]^T from telemetry y = [HR, SBP, DBP]^T.
- Dropout Resilience: Gracefully bypasses Kalman measurement update on NaN telemetry frames without state divergence.
- Isolation Forest Ensemble: Unsupervised secondary detector trained on residuals for multi-detector consensus.

INNOVATION & DETECTOR

Innovation Divergence Engine & Uncertainty Quantification

Robust Anomaly Detection with Persistence Filtering & Monte Carlo

- Mahalanobis Distance: D_M = sqrt(v^T * S^-1 * v), where v = y - y_pred and S is innovation covariance.
- Chi-Square Quantiles (k=3): Normal (<2.30, p>0.15) | Watch (2.30-3.14, p:0.02-0.15) | Significant (>=3.14, p<0.02).
- Persistence Filter (N=3): Requires divergence to persist for 3 consecutive steps, eliminating sensor artifact false alarms.
- 50-Twin Monte Carlo Ensemble: Draws 50 parameter samples from posterior N(x_hat, P) to generate 90% predictive bands [p5, p95].
- Reliability Guardrails: Low data reliability score automatically suspends predictions and warns clinician.

EXPERIMENTAL RESULTS

Cohort Validation & Empirical Outcomes

Rigorous Evaluation Across 30 Diverse Clinical Phenotypes

- Cohort Benchmark: 30 held-out synthetic patient profiles (10 Healthy, 10 Hypertensive, 10 Elderly).
- Predictive Accuracy: SBP MAE < 2.8 mmHg, DBP MAE < 1.9 mmHg, HR MAE < 1.2 bpm across normotensive cohort.
- Uncertainty Calibration: Empirical coverage of 90% Monte Carlo band measured at 89.4% (well-calibrated).
- Anomaly Detection Performance: 94.2% True Positive Rate with <3.1% False Alarm Rate on injected vascular crises.
- Multi-Detector Consensus: Divergence Engine achieved 1.4-step faster alarm response compared to standard Isolation Forest.

DELIVERABLES & ROADMAP

Demonstration Video, Deliverables & Roadmap

Public Artifacts, Unlisted Video Walkthrough & Future Work

- 15-20 Min Unlisted Demo Video: Comprehensive walkthrough covering all 8 tabs, 3D heart, and what-if simulation.
- Open-Access GitHub Repo: Complete source code, test suites, architecture diagrams, and presentation decks.
- Zero Permission Barriers: All documentation, slides, and code are publicly accessible under MIT License.
- Phase 2 Roadmap: Benchmark against PhysioNet PTB-XL & BIDMC human arterial datasets.
- Wearable Hardware Integration: Live Bluetooth Low Energy (BLE) streaming from smartwatch PPG sensors.